

# Introduction to Modeling

Attendance



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# Introduction to Modeling

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## Terms:

- **Model Training** - The process of estimating parameters used to make a prediction
- **Parameters** - Internal values of a machine learning model that are learned from the training data during the training process.
  - They define how the model transforms input data into the desired output.
- **Hyper-Parameters** - external configurations set before the training process begins.
  - They control how the training is conducted and influence the model's performance.

$$f(x) = w_1x_1 + \dots + w_nx_n$$

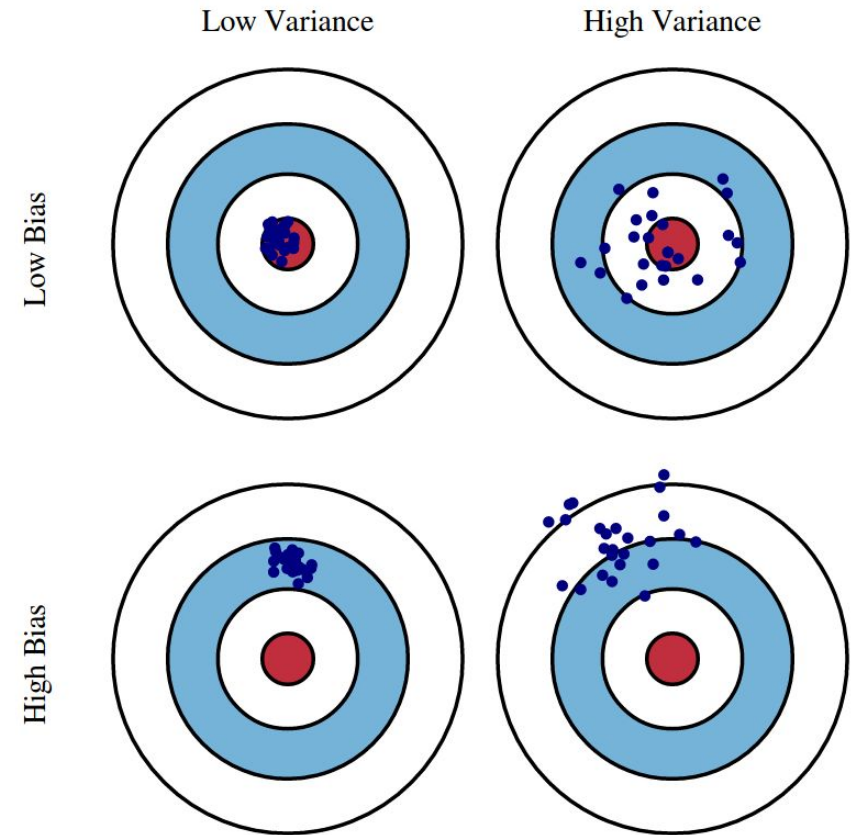
# Model Error

## breaking down error

- **total error** - how much the observed values differ from the predicted values

## three components to total error:

- **bias** - how much the model's prediction differs from the observed values due to the assumptions built into the model
- **variance** - a measure of how spread out predictions are from the model
- **irreducible error** - error inherent to the situation model, will always be there



# Model Error

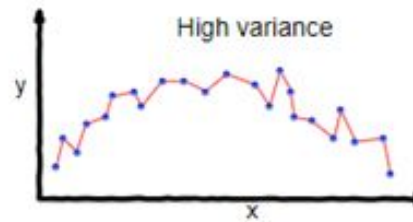
## when is a model underfit?

- a model is underfit if the model is too simple to fit the data well
  - contains high bias

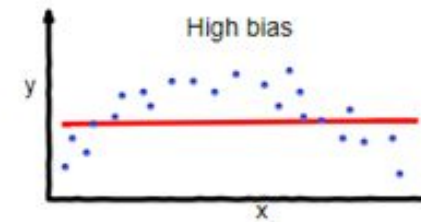
## when is a model overfit?

- a model is overfit if the model is too complex, fitting the data too closely and incorporating too much noise
  - contains high variance

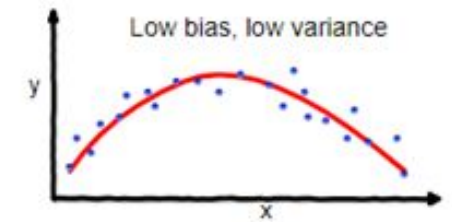
a model that is overfit or underfit does not generalize well to unseen data and should not be used



overfitting



underfitting

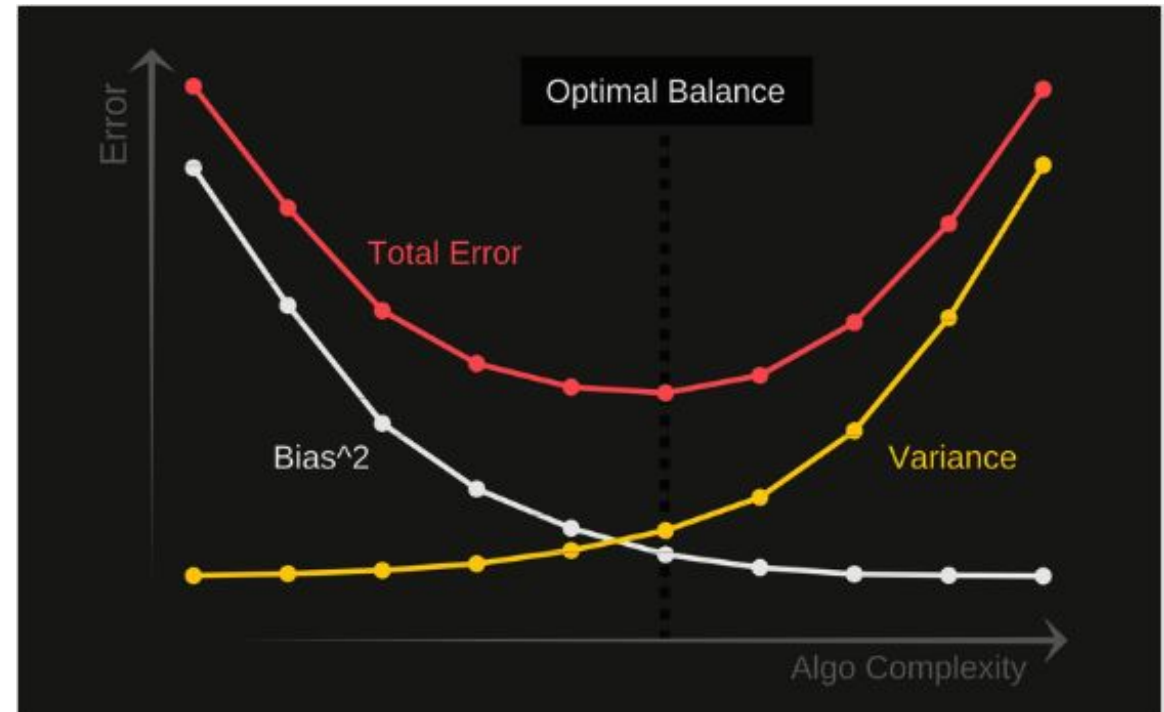


Good balance

# Bias-variance tradeoff

irreducible error is inherent to whatever situation is modeled, picking only a different model can only change the bias and the variance

- choosing a more complex model means the models predictions are closer to the observed sample values, which decreases the bias
- but, doing so will make the predictions more spread out
- an optimal model should be just complex enough to capture the general trend of the data without incorporating too much noise from the sample





# Q&A Session



# Attendance



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**Thursdays:** 11AM – 12PM & 2PM – 3PM

**Friday:** 3PM – 4PM

**Monday:** 12PM – 1PM



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